

Varenya Sri Mudumba

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Education

Texas A&M University, College Station, TX

Aug 2025 - May 2027

M.S. in Computer Science - GPA 4.0/4.0 (Expected May 2027)

National Institute of Technology, Durgapur

Dec 2020 - May 2024

B.Tech in Computer Science and Engineering - GPA 8.84/10.00

Technical Skills

Languages: Python, Rust, Go, C/C++, JavaScript, SQL, Bash.

AI Security: LLM red-teaming, prompt-injection defense, OWASP LLM Top 10, MITRE ATLAS, adversarial ML.

Frameworks & Governance: NIST AI RMF / CSF 2.0, STRIDE, RAG / agent security, secure MLOps.

Security: malware analysis, YARA, detection engineering, threat modeling, SAST / DAST, post-quantum crypto.

ML & Infra: PyTorch, LoRA / DoRA, RAG, MCP, Amazon Bedrock, ONNX, Docker, AWS, GCP, Git.

Experience

Research Intern

Jul 2026 - Aug 2026

Corelight

Remote, USA

- Built a self-supervised multi-agent system that reverse-engineers malware over a 23-tool static-analysis MCP (Ghidra, radare2, capa, FLOSS, androguard) to auto-generate per-family YARA rules, sandboxed from raw samples.
- Iteratively refined the agents' prompting and tool orchestration to raise detection coverage from **40-60% to 80-98%** and cut cost from **~\$10 to ~\$4** per family run, across a ~38K-sample corpus (138 families, 6 file types).

Graduate Researcher, Dr. Marcus Botacin's Lab

Feb 2026 - Present

Texas A&M University

College Station, TX

- Researching streaming algorithms for malware analysis to enable faster, more efficient YARA rule generation.
- Evaluating novel and hybrid detection methods, weighing detection accuracy against runtime and memory cost.

Software Development Engineer (Intern to SDE)

Feb 2024 - Aug 2025

Zscaler

India

- Integrated Zscaler Private Access (ZPA) zero-trust secure-access features adopted by 7+ internal teams.
- Built a Pytest platform running 900+ daily regression tests, and cut environment setup from 18 hours to 3 by automating AWS/GCP provisioning with Terraform and Ansible, catching regressions within hours of a commit.

Projects

HMCP - Post-Quantum Secure MCP Middleware

Feb - May 2026

- Architected quantum-safe MCP middleware in Rust (gRPC transport) with a Kyber-768/X25519 hybrid KEM over a Noise XX handshake, establishing forward-secret, tamper-evident channels between agents and tools.
- Added a 3-tier input sanitizer (homoglyph, mixed-script, control-char), plus DID authentication and rate limiting, to block prompt-injection, spoofing, replay, and man-in-the-middle attacks across the transport.

Poisoned Pipelines - Agentic-AI & MCP Threat Model

Apr 2026

- Authored a threat taxonomy and defense framework for agentic AI coding tools and the Model Context Protocol; defined **seven threat classes** mapped to OWASP LLM Top 10 and MITRE ATLAS with concrete attack scenarios.
- Mapped **17 disclosed CVEs** with STRIDE analysis and layered, defense-in-depth mitigations across the taxonomy.

ZTBI - Zero-Trust Browser Interface

Nov - Dec 2025

- Built a Chrome extension running DistilBERT on-device (ONNX) to detect prompt injection and PII leaks in real time against untrusted page content, with all inference kept on-device and no data leaving the browser.
- Hardened it against tag-splitting, indirect-injection, and asynchronous exfiltration channels.

Additional

- **Publication:** "PSO-Driven Test Pattern Generation for Hardware Trojan Detection," IEEE 2024. DOI.
- 7 vulnerabilities discovered and disclosed to vendors; merged 2 PRs into IBM's qiskit-machine-learning.
- Competitive programming: peak CodeChef rating 1893; 500+ algorithmic problems solved.
- **Work Authorization:** F-1 OPT (STEM) - up to 36 months, no immediate sponsorship.